Chapter 7: Kinetic Energy

The following subjects will be discussed

- Energy in General.
- Kinetic Energy.
- Work.
- Power.

Introduction

What is Energy?

Why do we need the "energy" concept?

Forces behind an action cannot always be identified

Energy is a "book-keeping" quantity. Everything is accounted for.

Conservation laws are universal.

Energy is scalar

Energy has many forms that fit all situations.

Examples: electrical energy, thermal energy, nuclear energy, …

The every-day usage of the word is different.

Units of Energy: Joules (J). Basic units depend on the type of energy.

How to "sense" energy?

Energy can be classified into two groups: Kinetic energy (motion) and potential energy (no motion).

Kinetic Energy

In this course, we are concerned with the kinetic and potential energies associated with a macroscopic particle or a group of particles (a system).

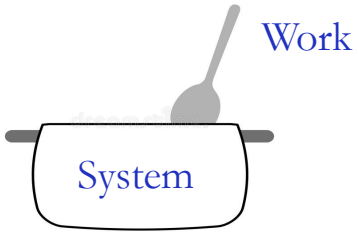
An isolated system is a system where no exchange of energy takes place with the outside of the system.

Kinetic energy is the type of energy associated with motion ($v = 0 \rightarrow KE = 0$). It is calculated at a specific velocity.

Kinetic energy can be related directly or indirectly to "some" force. Using the equations of motion $v^2 = 2ax$, we get $KE = \frac{1}{2}mv^2$.

Potential energy is the type of energy associated with position ($x = 0 \rightarrow PE = 0$). It is calculated at a specific position. It can also be looked at as a "stored" energy.

Work is the mechanism by which energy is either added to (positive work) or taken from (negative work) a given system. Usually associated with some force. $W = \vec{F} \cdot \vec{d}$



Work

The change in the system's kinetic energy is due to Work.

$$KE(final) - KE(initial) = Work$$

$$Work = KE_f - KE_i = \Delta KE$$

$$v_f^2 - v_i^2 = 2ax \quad \rightarrow \quad \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = m\vec{a}\cdot\vec{x} = \vec{F}\cdot\vec{x} = W$$

Work by gravitational force (No external force: free fall example)

Going up:

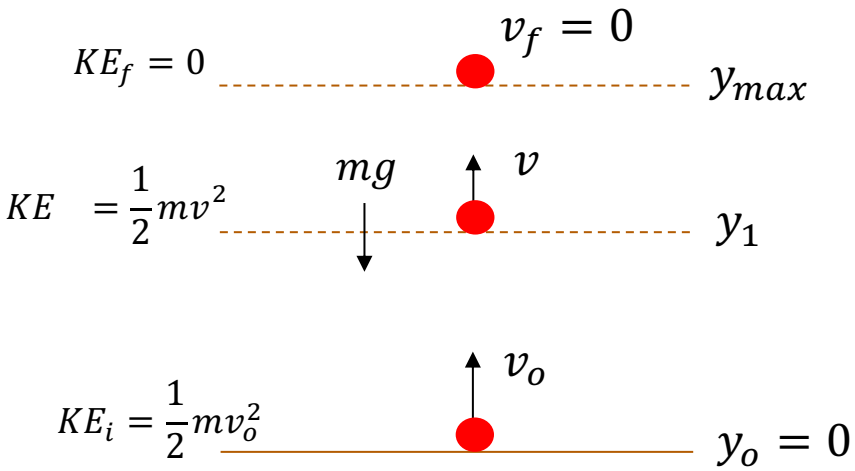
$$W = \vec{F}\cdot\vec{d} = mg\,(y_{max} - y_o)\cos\phi$$

$$W = 0 - \frac{1}{2}mv_o^2$$

Going down:

$$W = \vec{F}\cdot\vec{d} = mg\,(y_{max} - y_o)\cos\phi$$

$$W = \frac{1}{2}mv_o^2 - 0$$



Work by Gravitational Force

Work by gravitational force (With external force: elevator example)

$$Work = KE_f - KE_i = \Delta KE = 0$$

Going up:

Gravity $\quad W_g = \vec{F} \cdot \vec{d} = mg\,(y_f - y_o)\cos\phi$ $\quad -1$

Cable $\quad W_c = \vec{F} \cdot \vec{d} = T\,(y_f - y_o)\cos\phi$ $\quad +1$

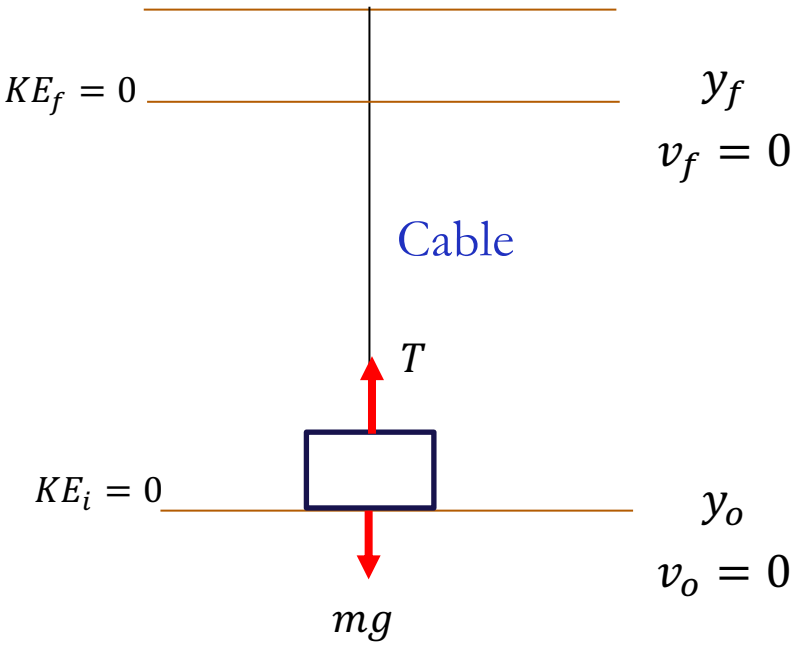
Total $\quad W_g + W_c = 0$

Going down:

Gravity $\quad W_g = \vec{F} \cdot \vec{d} = mg\,(y_f - y_o)\cos\phi$ $\quad +1$

Cable $\quad W_c = \vec{F} \cdot \vec{d} = T\,(y_f - y_o)\cos\phi$ $\quad -1$

Total $\quad W_g + W_c = 0$



Examples

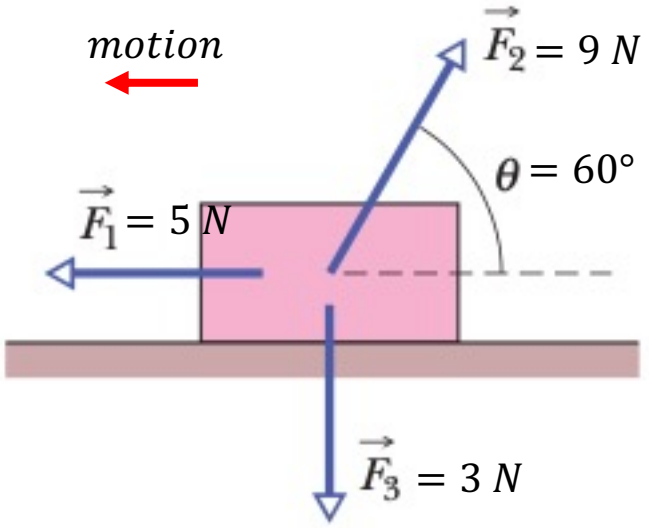
1) In response to the three forces, the block has moved 3 meters to the left. How much net work did the forces apply on the block during movement. Did the KE increase or decrease?

For a constant force

$$W = \vec{F} \cdot \vec{x}$$

$$W_{net} = F_1\, d \cos\phi + (F_2 \cos 60)\, d \cos\phi$$

$$W_{net} = (5)(3)(1) + (9)(\cos 60)\ (3)(-1) = 1.5\ J$$



Positive net work → KE increases

Examples

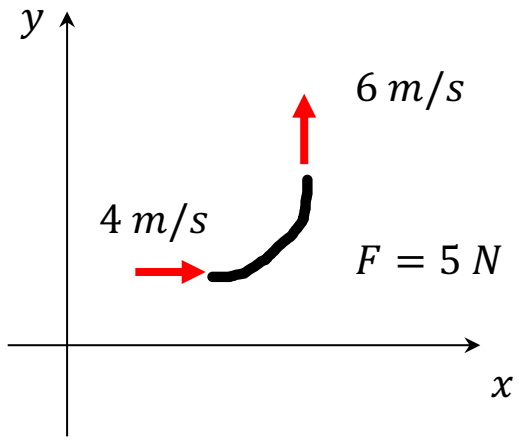
2) The object has changed course and speed by a 5-N force. How much work did the force apply on the object?

$m = 2\ kg$

$$W = \Delta KE$$

$$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$W = \frac{1}{2}(2)(6^2 - 4^2) = 20\ J$$



Examples

3) A force $F_r = 50\ N$ is applied on the block as it slides down by 0.5 meters. The block gains 80 Joules of energy as it slides the 0.5 meters. How much energy would the block gain if no force was applied? Also, find the mass of the block.

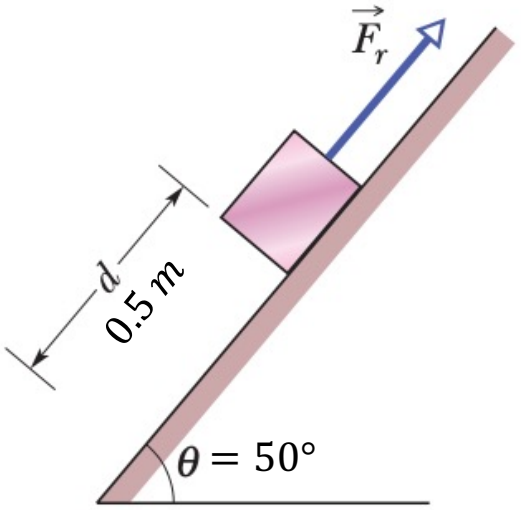
With the force.

$$W_g + W_F = \Delta KE = 80\ J$$

$$W_F = F_r\, d \cos\phi = (50)(0.5)(-1) = -25\ J$$

$$W_g = 80 - (-25) = 105\ J$$

$$F_g = W_g/(d \cos\phi) = 105/(0.5)(1) = 210\ N$$



Without the force.

$$W_g = \Delta KE = F_g\, d \cos\phi = (210)(0.5)(1) = 105\ N$$

$$m = \frac{210}{g \sin 50} = 28\ kg$$

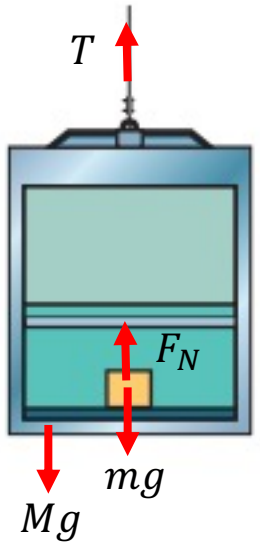
Examples

4) Find the work done by the cable as the elevator goes up 2.4 meters.

$$F_N = 3\ N$$

$$m = \frac{1}{4}\ kg$$

$$M = 900\ kg$$



Newton's Law

$$F_N - mg = ma \quad \Rightarrow \quad a = \frac{F_N - mg}{m}$$

$$T - (m+M)g = (m+M)a \quad \Rightarrow \quad T = (m+M)(g+a)$$

$$W_T = T\,d = 25.9\ kJ$$

Work by a Spring

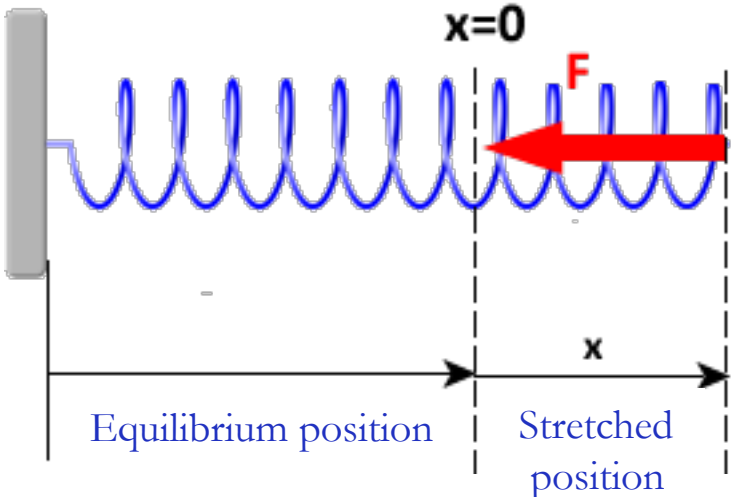
The "restoring" force in a spring is similar in nature to gravitational force, but with one major difference; it's a variable force.

$$F_s = -k\,x$$

$$W_s = \int_{x_i}^{x_f} F_s\,dx \quad = -\int_{x_i}^{x_f} (k\,x)\,dx$$

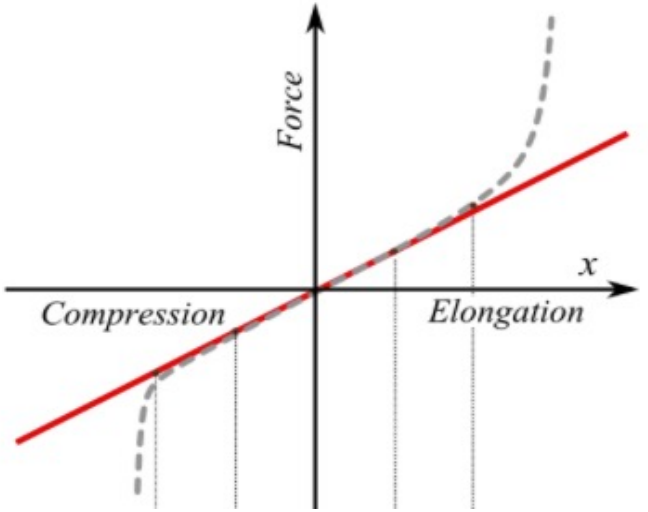
$$W_s = -\left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right)$$

$$W_s = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$



With only spring force $W_s = KE_f - KE_i = \Delta KE$

With other forces $W_s + W_F = \Delta KE$

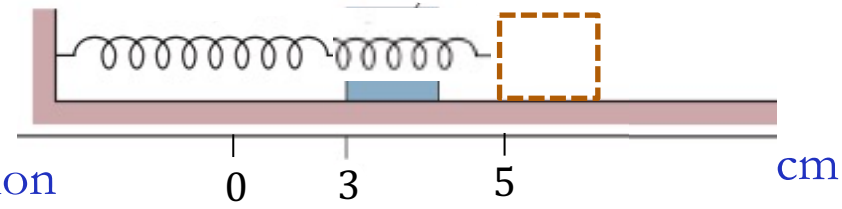


Examples

1) The spring is initially at rest then moved to +3 cm position. The work curves shows this part of the motion.
Then the block is moved to +5 position and released.

From the curve we find k.

$$W_s = \frac{1}{2}kx^2 \rightarrow k = \frac{(2)(4)}{0.02^2} = 2\times10^4\ N/m$$

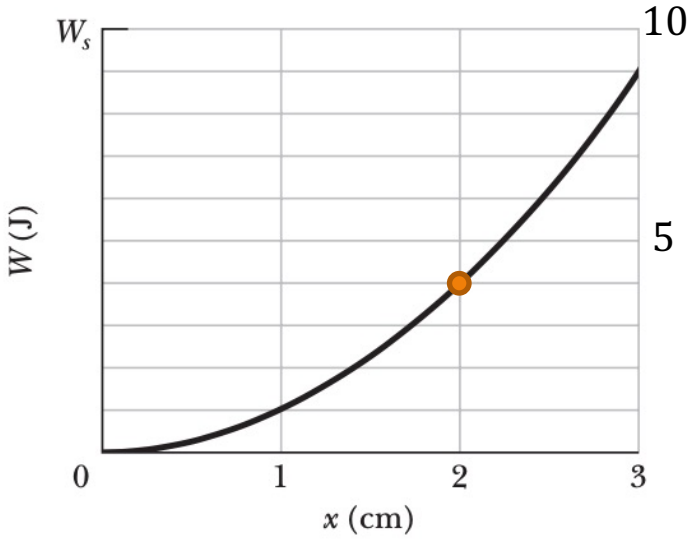


Find the work going from +5 to +4 cm position

$$W_s = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2 = \frac{1}{2}k\,5^2 - \frac{1}{2}k\,4^2 = 9\,J$$

Find the work going from +5 to -2 cm position

$$W_s = \frac{1}{2}k\,(5^2 - 0^2) + \frac{1}{2}k\,(0^2 - (-2)^2) = 21\,J$$



Find the work going from +5 to -5 cm position

$$W_s = 0$$

Examples

2) The force acting on an object moving along the x axis is given in N by

$$F = (cx - 3x^2)\vec{i}$$

At $x = 0$, the kinetic energy of the object is 20 J, at $x = 3$ m, it is 11 J. Find c.

$$Work = KE_f - KE_i = \Delta KE = 11 - 20 = -9\,J$$

Also

$$W = \int_{x_i}^{x_f} F\,dx = \int_{x_i}^{x_f} (cx - 3x^2)\,dx = \left(\frac{cx^2}{2} - x^3\right)\Big|_0^3 = \frac{9}{2}c - 27$$

$$\frac{9}{2}c - 27 = -9 \;\rightarrow c = 4\ N/m$$

Power

Power is the rate at which work is done, with units of J/s or Watt.

Average power $P_{ave} = \frac{\Delta W}{\Delta t}$

Instantaneous power $P = \frac{dW}{dt}$

What does it mean when work is constant with time?

Other units of power

Horsepower $1\ hp = 746\ W$

KiloWatt hour is energy

1 kW-hr = 3.6×10^6 J

Tesla S3 model Battery 75 kW-hr

Note the following.

$$P = \frac{dW}{dt} = \frac{d(\vec{F} \cdot \vec{x})}{dt} = \vec{F} \cdot \frac{d(\vec{x})}{dt} = \vec{F} \cdot \vec{v}$$

Examples

1) A machine carries a 4-kg object from one position d_i to a final position d_f with a force F in 12 seconds, as given by:

$$\vec{d}_i = 0.5\,\vec{\imath} + 0.75\,\vec{\jmath} + 0.2\,\vec{k} \quad \text{m}$$

$$\vec{d}_f = 7.5\,\vec{\imath} + 12\,\vec{\jmath} + 7.2\,\vec{k} \quad \text{m}$$

$$\vec{F} = 2\,\vec{\imath} + 4\,\vec{\jmath} + 6\,\vec{k} \quad \text{N}$$

Work done

$$W = \vec{F} \cdot \vec{d} = \vec{F} \cdot (\vec{d}_f - \vec{d}_i)$$

$$W = \left(2\,\vec{\imath} + 4\,\vec{\jmath} + 6\,\vec{k}\right) \cdot \left(7\,\vec{\imath} + 11.25\,\vec{\jmath} + 7\,\vec{k}\right) = 101\,J$$

Average power

$$P_{ave} = \frac{\Delta W}{\Delta t} = \frac{101}{12} = 8.4\,W$$

Examples

2) A 5-N force is acting on a 15-kg object initially at rest. Find the work done on the object the first three seconds. Also, calculate the instantaneous power and the average power at the end of the third second.



To find the work we need the distance.

It is a constant force, so constant acceleration. Use the equations of motion.

$$x = \frac{1}{2} a\, t^2 \qquad a = \frac{F}{m} = \frac{1}{3} \; m/s^2$$

$t = 1\ s$	$x = \frac{1}{6}$ m	$d = \frac{1}{6}$ m	$W = \frac{5}{6} = 0.83\ J$
$t = 2\ s$	$x = \frac{2}{3}$ m	$d = \frac{1}{2}$ m	$W = \frac{5}{2} = 2.5\ J$
$t = 3\ s$	$x = \frac{3}{2}$ m	$d = \frac{5}{6}$ m	$W = \frac{25}{6} = 4.17\ J$

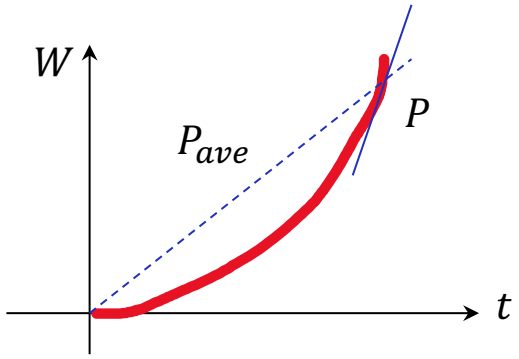
Instantaneous power

$$P = \frac{dW}{dt} = F\frac{dx}{dt} = F\ at$$

$$P = 5\ W$$

Average power

$$P_{ave} = \frac{\Delta W}{\Delta t} = \frac{7.5}{3} = 2.5\ W$$



Q & A

Send me an email if
you have any
questions.